

Thermodynamic State Vector Inventory Discrepancy Evaluator Core Helper: Ensuring Mass Conservation and Thermodynamic Consistency in Biogeochemical Simulations

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Abstract

Complex biogeochemical simulations modelling Earth pod dynamics require rigorous adherence to mass conservation (First Law of Thermodynamics) and bounded energy dissipation (Second Law of Thermodynamics). Cumulative floating-point drift and numerical errors during multi-reservoir elemental cycling can compromise simulation fidelity. This paper presents the architectural and mathematical design of Sprint 068: the **Thermodynamic State Vector Inventory Discrepancy Evaluator Core Helper** (`src/thermodynamics/state_validator.ts`). By establishing isolated, deterministic comparison routines against per-element tolerances (τ_i), the `StateValidator` provides real-time verification of thermodynamic state vectors S_{actual} versus expected baselines S_{expected} . We formalize the stock transfer equations, discrepancy metrics, and validation predicates, demonstrating how automated state inspection preserves thermodynamic integrity across simulated ecological monads.

Keywords: Thermodynamic state vectors, mass conservation, exergy dissipation, biogeochemical modeling, numerical drift, systems ecology.

1 Introduction and Systems Ecology Context

The Web of Life simulation models continuous biogeochemical cycles—specifically Carbon, Nitrogen, Phosphorus, and Water—driven by solar energy influx and constrained by closed-system mass conservation principles. Within this architecture, ecological monads execute discrete state transitions, transferring matter between environmental reservoirs and living biomass pools.

However, computational modelling of continuous-time ecological dynamics introduces numerical artifacts, notably floating-point rounding errors and minor mass leakage during matrix transformations. Left unchecked, these errors compound over simulation epochs, violating the foundational conservation laws of systems ecology.

To address this, Sprint 068 introduces the `StateValidator` core helper (`src/thermodynamics/state_validator.ts`). This module acts as an independent auditing agent that verifies system state vectors against strict elemental tolerances without mutating underlying simulation data. Official repository: <https://github.com/pascalranoroarijaona/WebOfLife>.

2 Mathematical Formalization

Let a thermodynamic state vector S be defined as a mapping of elemental inventories and energy stocks:

$$S = \{e_i \mid i \in \{\text{carbon, nitrogen, phosphorus, water, energy, } \dots\}\} \quad (1)$$

Given an actual state vector S_{actual} resulting from monad execution and an expected baseline state vector S_{expected} derived from stoichiometric conservation laws, the absolute discrepancy Δ_i for each element i is evaluated as:

$$\Delta_i = |e_{i,\text{actual}} - e_{i,\text{expected}}| \quad (2)$$

Each element possesses a configurable tolerance threshold τ_i , defaulting to $\epsilon = 10^{-6}$ for mass pools and $\epsilon = 10^{-4}$ for energy. The elemental validation predicate P_i is expressed as:

$$P_i = \begin{cases} \text{true}, & \text{if } \Delta_i \leq \tau_i \\ \text{false}, & \text{if } \Delta_i > \tau_i \end{cases} \quad (3)$$

The global system validity (`isValid`) is established by the logical conjunction of all element predicates:

$$\text{isValid} = \bigwedge_i P_i \quad (4)$$

Furthermore, the maximum system discrepancy Δ_{max} across all tracked pools is computed to monitor numerical drift trajectories:

$$\Delta_{\text{max}} = \max_i(\Delta_i) \quad (5)$$

3 Thermodynamic Guarantees

1. **First Law Enforcement:** By enforcing strict absolute difference checks against τ_i , the validator ensures that biogeochemical cycles maintain mass balance across carbon, nitrogen, phosphorus, and water reservoirs. Any unaccounted creation or destruction of matter immediately flags `isValid: false`.
2. **Second Law Bounds:** Energy discrepancies track allowable thermodynamic dissipation limits, preventing perpetual motion artifacts or uncalibrated free energy generation within simulated Earth pods.

4 Conclusion

The implementation of `StateValidator` provides simulation runners with robust diagnostic capabilities, safeguarding ecological models against numerical drift and ensuring strict adherence to thermodynamic laws.